

The Modulus: a Machine-Checked Operator Bound on the Fluctuation Sector of the Coupled $\mathbb{Z}_2$ Slice

one construction, two debts — and the one it does not pay

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Abstract

Two companion papers were left carrying the same debt from opposite sides. The gap paper proved that every eigenvalue of the coupled slice other than the Perron eigenvalue is *strictly* smaller in modulus, and said plainly that this provides *no modulus of separation* [2]. The bridge paper proved that the Gibbs correlations of the spatial system are matrix elements of a self-adjoint transfer operator, and obtained geometric decay only *under a contraction hypothesis it did not discharge* [3]. The missing step is identical in both: finitely many strict inequalities are not an operator-norm bound.

Proved. We construct specGap , the largest $|\mu|$ over the eigenvalues different from the Perron eigenvalue λ , and prove $\text{specGap} < \lambda$. We then prove the operator bound: for every observable u orthogonal to the Perron vector, $\|Ku\| \leq \text{specGap} \cdot \|u\|$. That is exactly the hypothesis the bridge paper carried, so its bound becomes unconditional.

And a warning we state before anyone else has to. $\text{specGap} < \lambda$ is *not* $\text{specGap} < 1$: the kernel is unnormalised, so both are typically far above one and specGap^N *grows*. On its own the unnormalised bound therefore controls growth, it does not exhibit decay. The rate that is below one is the *relative* one, $\text{specRatio} = \text{specGap}/\lambda < 1$, and the decay statement is that the fluctuation contribution is suppressed by specRatio^N *relative to the Perron scale* λ^N . Both forms are proved; only the second is called decay.

The step that does not follow from the inequalities is the one about eigenvectors *at* λ : geometric simplicity, proved in the Perron paper for an arbitrary eigenvector rather than a positive one [1], makes them *invisible* to a fluctuation observable, so the top term of the spectral sum vanishes instead of merely being bounded.

Not proved. specGap *depends on the extent*, and nothing here bounds it away from λ uniformly. Direct diagonalisation gives $\text{specGap}/\lambda = 0.9205, 0.9829, 0.9964, 0.9992$ at $L = 2, 3, 4, 5$ for one parameter pair: a geometric bound whose rate tends to 1 is empty in the limit, and that is reported, not hidden. The relative decay is still not the normalised Gibbs *expectation*: that needs a lower bound on the partition function, which is not proved here. Reflection positivity is untouched, and nothing in this paper is a claim about $\text{SU}(N)$, the continuum limit, or the Yang–Mills mass gap.

1 The same debt, from two sides

The gap paper’s headline is a strict separation without a modulus. The bridge paper’s decay theorem carries a contraction hypothesis. It is worth stating precisely why these are one debt and not two.

A strict inequality $|\mu| < \lambda$ for each of finitely many eigenvalues does give $\max |\mu| < \lambda$ — taking a maximum over a finite set is not the difficulty. The difficulty is that $\max |\mu|$ over

the non-Perron eigenvalues is not, by itself, known to *dominate the operator* on the orthogonal complement of the Perron vector. Getting from one to the other requires the spectral decomposition: an orthonormal basis of eigenvectors, so that an observable can be expanded and its image measured coordinate by coordinate.

That is what this paper supplies, and it pays both debts with one construction. We prove the inequality $\|Ku\| \leq \text{specGap}\|u\|$; we do *not* prove that specGap *equals* the restricted operator norm. The numerical section reports that the bound is attained, but attainment is **verified**, not proved, and no theorem here depends on it.

2 The construction

Let K be a strictly positive symmetric kernel on a finite nonempty type, with Perron data (v, λ) : $v > 0$ pointwise and $Kv = \lambda v$.

Definition 2.1. Let μ_j be the eigenvalues of K and set

$$\text{specGap} = \max_j \begin{cases} 0 & \text{if } \mu_j = \lambda, \\ |\mu_j| & \text{otherwise.} \end{cases}$$

[specGap](#). Eigenvalues equal to λ contribute 0 rather than λ . That is legitimate — and it is the whole content of §3.

Theorem 2.2 (the modulus). $\text{specGap} < \lambda$.

[specGap_lt](#). Each term of the maximum is either $0 < \lambda$, or $|\mu_j| < \lambda$ by the gap paper applied to the j -th basis vector, which is a nonzero eigenvector because it has unit norm.

3 Why eigenvectors at the top may be discarded

Theorem 3.1. *If $\mu_j = \lambda$ and u is orthogonal to the Perron vector, then u has no component along the j -th eigenvector.*

[inner_specBasis_eq_zero_of_top](#).

Remark 3.2. *This is the step that does not come from the inequalities, and it is why Definition 2.1 may assign 0 to the top of the spectrum. The Perron paper proves geometric simplicity for an arbitrary eigenvector with eigenvalue λ , not merely a positive one [1]; hence the j -th basis vector is a multiple of v , and orthogonality to v kills the coordinate outright. Had only the positive case been available, the top term could have been bounded but not removed, and the maximum in Definition 2.1 would have had to include λ — making Theorem 2.2 false and the whole construction vacuous.*

4 The operator bound

Theorem 4.1 (the bound). *For every u with $\langle v, u \rangle = 0$,*

$$\|Ku\| \leq \text{specGap} \cdot \|u\|.$$

`norm_act_le_specGap`.

The proof is the expansion in the eigenbasis. Writing c_j for the coordinates of u , the coordinates of Ku are $\mu_j c_j$ (`inner_specBasis_act`), so by Parseval (`eucNorm_sq_eq_sum`)

$$\|Ku\|^2 = \sum_j \mu_j^2 c_j^2, \quad \|u\|^2 = \sum_j c_j^2,$$

and each term is dominated: if $\mu_j = \lambda$ then $c_j = 0$ by Theorem 3.1, and otherwise $|\mu_j| \leq \text{specGap}$ by definition.

5 The endpoint

The bridge paper's fluctuation sector is invariant under a kernel that *fixes* the vacuum. Here the kernel is unnormalised and the vacuum is an eigenvector with eigenvalue λ , so invariance is reproved in that form (`perp_invariant_eigen`); the scalar drops out because the pairing is bilinear. Iterating gives `iterate_norm_le_specGap`.

Theorem 5.1 (the bridge bound, without its hypothesis). *For the coupled spatial system, at every finite extent L , every β , and every strictly positive source weight w : if the dressing of A is orthogonal to the Perron vector, then for every N*

$$\left| \sum_X A(X_0) A(X_N) \mathcal{W}(X) \right| \leq \text{specGap}^N \|\sqrt{w}A\|^2, \quad \text{specGap} < \lambda.$$

`gibbs_pathSum_bound_unconditional`.

Remark 5.2 (why that is not yet decay). $\text{specGap} < \lambda$ *does not give* $\text{specGap} < 1$. The kernel is unnormalised: in the probe of §7, $L = 2$ with $\beta = 0.8$, $\gamma = 1.2$ gives $\lambda \approx 56.89$ and $\text{specGap} \approx 52.37$, so specGap^N grows. Theorem 5.1 is a sub-Perron growth bound. The decay is Theorem 5.3.

Theorem 5.3 (the decay statement). *With $\text{specRatio} = \text{specGap}/\lambda$, under the same hypotheses,*

$$\left| \sum_X A(X_0) A(X_N) \mathcal{W}(X) \right| \leq \text{specRatio}^N \left(\lambda^N \|\sqrt{w}A\|^2 \right), \quad \text{specRatio} < 1.$$

Equivalently, the normalised kernel $\lambda^{-1}K$ contracts the fluctuation sector by $\text{specRatio} < 1$.

`gibbs_pathSum_relative_decay`, `specRatio_lt_one`, `norm_act_normalizedKernel_le`.

Remark 5.4 (non-vacuity). *Dressing is multiplication by a strictly positive function, so every fluctuation vector is the dressing of an observable: whenever the configuration space has two points there is a nonzero A satisfying the hypothesis, and Theorem 5.1 is not quantified over an empty set (`exists_nonzero_dressed_fluctuation`).*

6 Scope: three things this does not give

Nothing uniform in the extent. This is the substantive limitation. specGap is built from the spectrum at a fixed extent and depends on it. Direct dense diagonalisation gives, for $\beta = 0.8$, $\gamma = 1.2$,

$$\text{specGap}/\lambda = 0.6640, 0.9205, 0.9829, 0.9964, 0.9992 \quad \text{at } L = 1, \dots, 5,$$

and for $\beta = 0.3$, $\gamma = 0.4$,

0.2913, 0.4665, 0.5518, 0.5938, 0.6159.

The first family collapses towards 1, where a bound of the form specGap^N says nothing in the limit. These numbers are **verified** and **not proved**, no theorem depends on them, and they are printed because a paper that reported only Theorem 5.1 would be reporting the half that does not settle the question.

Relative decay, not the Gibbs expectation. Theorem 5.3 compares with λ^N , which is the growth scale of the partition function but is *not* the partition function. Turning it into a statement about the normalised expectation needs a lower bound on that partition function of order λ^N — which needs the index of the top eigenvalue identified and its overlap with the dressed constant observable bounded below. Neither is done here, and the word *clustering* is deliberately not used.

A bound, not an identification. We prove $\|Ku\| \leq \text{specGap}\|u\|$ on the fluctuation sector. We do not prove that specGap equals the restricted operator norm; that would need a nonzero vector attaining it, hence the index of the top eigenvalue, which is the same missing ingredient as in the previous paragraph. The probe reports attainment as **verified**.

No claim about the physical target. Reflection positivity is untouched, and nothing here concerns $\text{SU}(N)$, the continuum limit, or the Clay problem [10]. The distance to that problem is unchanged.

7 What is verified and not proved

A machine-checked proof guarantees that the definitions are internally consistent, not that they name the object the prose claims. The construction was therefore recomputed by direct dense diagonalisation, by a script with no access to the formalisation. Three things were checked: that $\text{specGap} < \lambda$ in every case; that $\|Ku\| \leq \text{specGap}\|u\|$ on 400 random fluctuation vectors per case; and that the bound is *attained* by the subdominant eigenvector, so it is sharp rather than a loose over-estimate. All cases pass.

One consequence is worth recording. The measured $\text{specGap}/\lambda$ reproduces, digit for digit, the subdominant ratios the gap paper reported as measured and unnamed [2]. The object constructed here is that number.

8 Formalisation notes

The library had the hard half. `mathlib` provides an orthonormal eigenbasis and real eigenvalues for a Hermitian matrix in finite dimension. The work in this module is not the spectral theorem; it is the bridging — writing the kernel as a Hermitian matrix, matching `mathlib`’s eigen-equation to the form the companion papers state, and translating norms and pairings between the two spellings. That bridging is where the effort went, and it is the usual place.

Where the unnormalised kernel forced a reproof. The bridge module’s invariance lemma assumes the vacuum is *fixed*. Its kernel is unnormalised, so the vacuum is an eigenvector with eigenvalue $\lambda \neq 1$ in general, and the lemma does not apply verbatim. Rather than reopen a frozen module, the invariance is reproved here in the eigenvector form. It is three lines, because the eigenvalue factors out of a bilinear pairing.

9 Reproducibility

All artifacts are at source checkpoint `7f3c5add2aa78757203c3dc74758fcd6233df83d` on branch `main`. Every hyperlink was resolved against that commit, and checked to point at an actual declaration line, before compilation.

Quantity	Value
Full core build	8430 jobs, success
Oracle commands	2655
with axiom dependencies	2632
axiom-free	23
Distinct axioms across all outputs	<code>propext</code> , <code>Quot.sound</code> , <code>Classical.choice</code>
Occurrences of <code>sorryAx</code>	0
Project axioms	0
Errors	0

The module is [YangMills/OS/SpatialSpectral.lean](#); the probe is [scripts/probe_spatial_spectral.py](#); verdicts are in [docs/VERIFICATION-LEDGER.md](#).

References

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